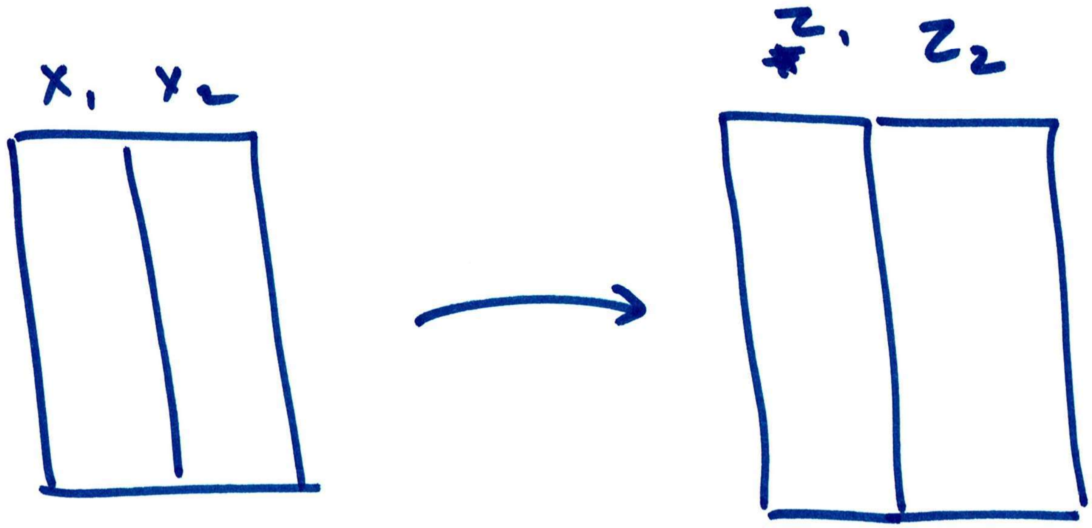
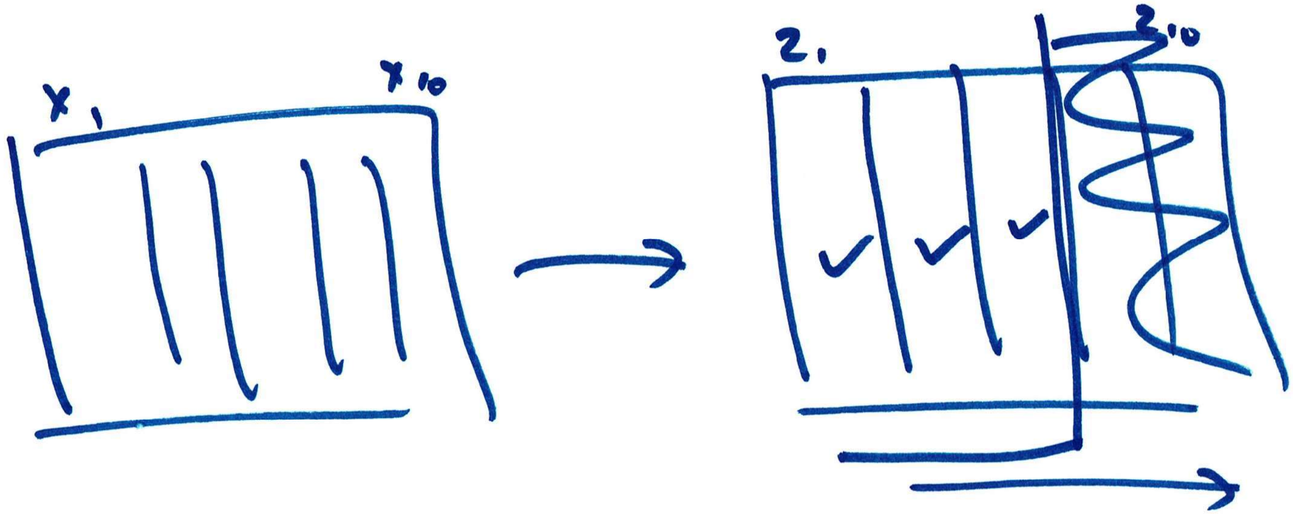
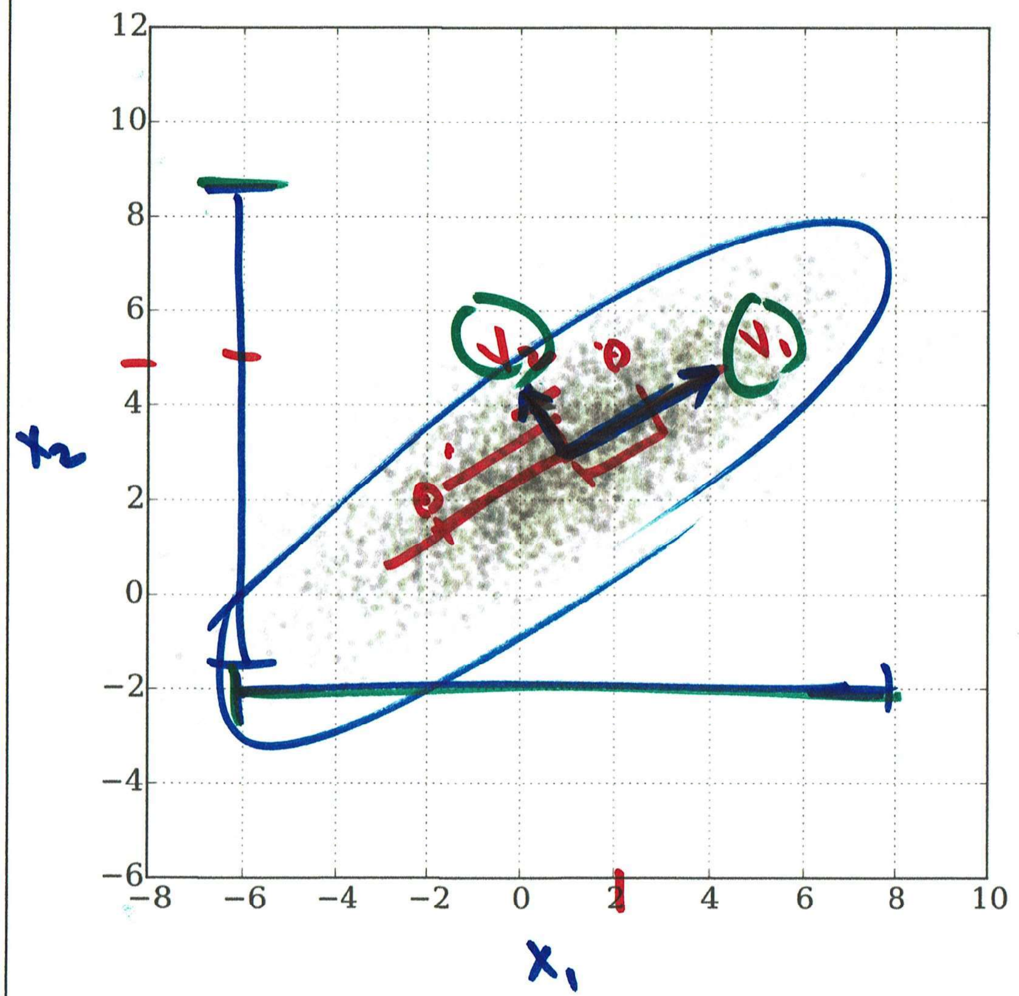
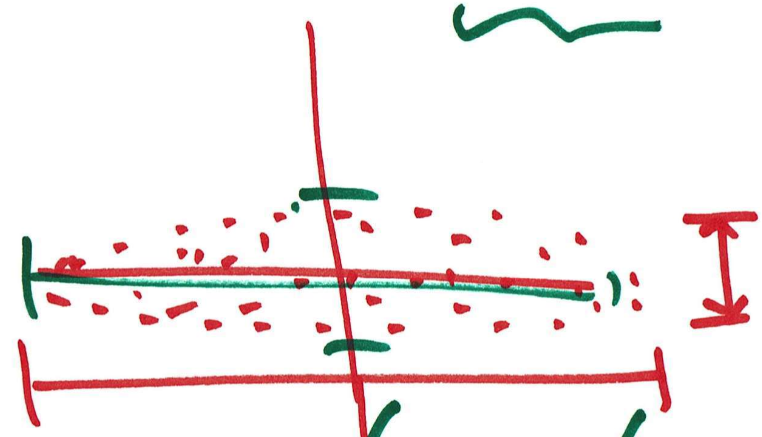

x1
x2
z1
z2


x 1
x 10
z 1
z 10



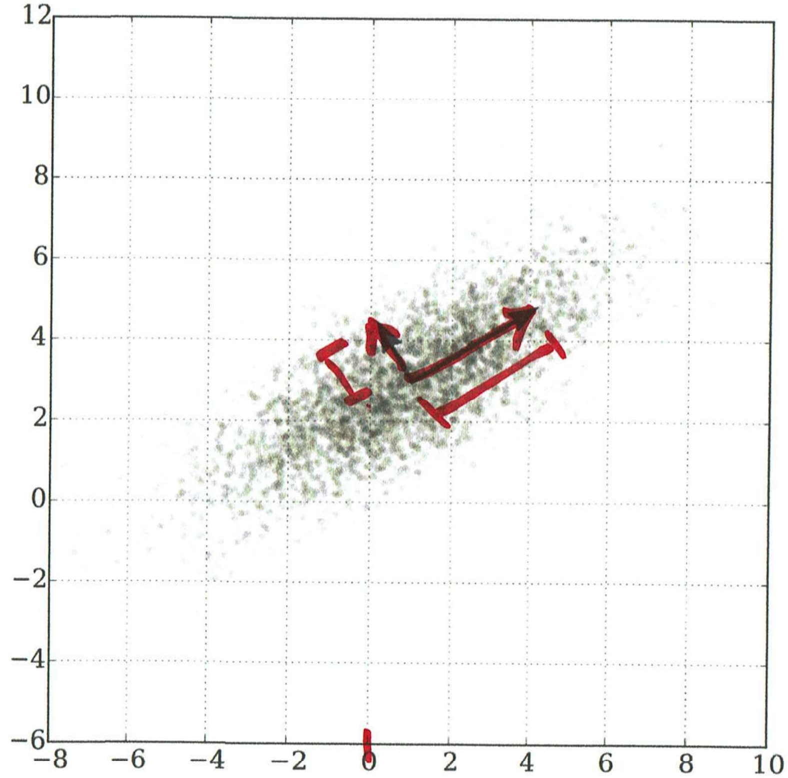
$$\begin{array}{cc} x_1 & x_2 \\ 2 & 5 \\ -2 & 2 \\ 1.2 & 0.8 \end{array} \rightarrow \begin{array}{cc} z_1 & z_2 \\ 1 & 0.8 \\ -2 & 0.5 \\ 1.9 & 0.1 \end{array}$$

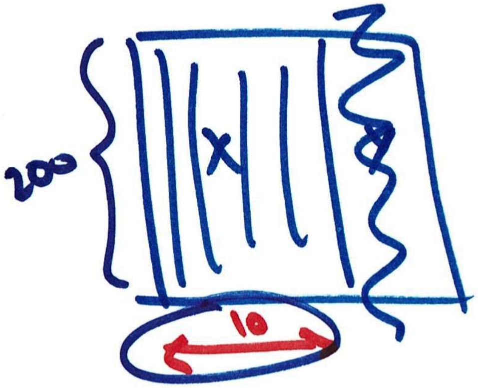


$$\begin{cases} Z_1 = \square\, x_1 + \square\, x_2 \\ Z_2 = \square\, x_1 + \square\, x_2 \end{cases}$$

PCA

- creates new variables using linear combinations of old variables
- is designed to create variables that are independent of one another
- also manages to tell us how important each of these new variables are
- this “importance”, helps us to choose how many variables we will use





$$X_{new} = \frac{X - \mu}{\sigma}$$

$$\downarrow$$

$$C = \mathrm{Cov}(X_{new})$$

$$\downarrow$$

Eigen decomposition

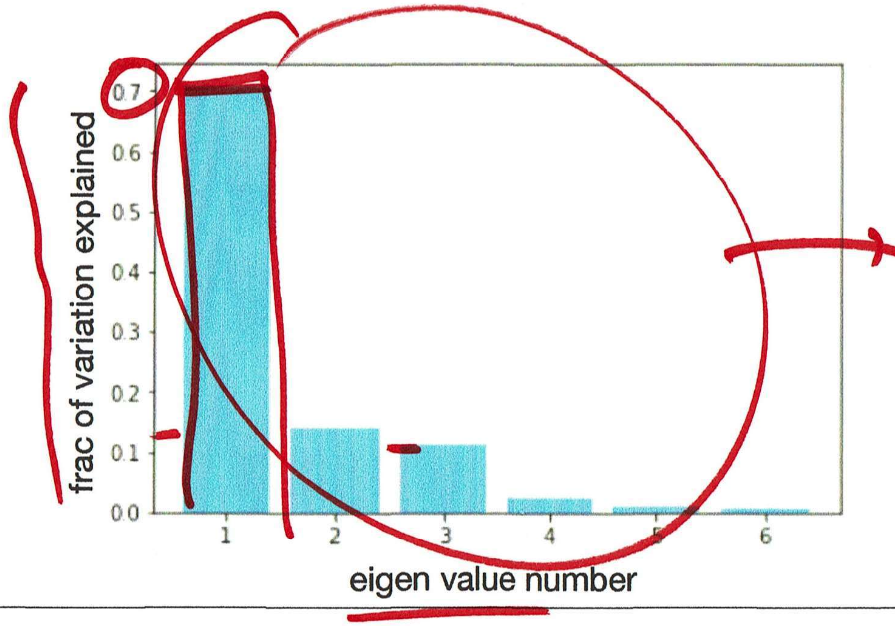
$$\begin{matrix} e_1, & e_2, & \ldots & \ldots & e_{10} \\ v_1, & v_2 & \ldots & \ldots & v_{10} \end{matrix}$$

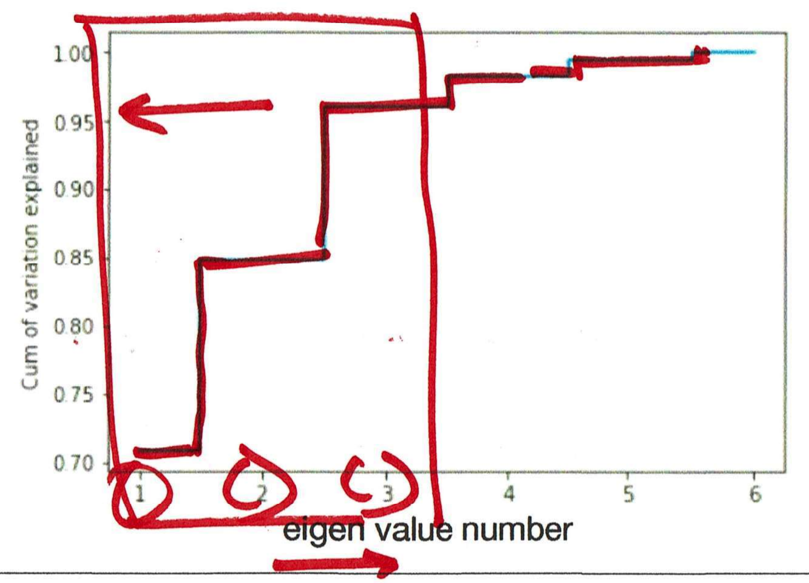
$$\boxed{Z} \quad \boxed{10 \times 3} \leftarrow W = \begin{array}{c} \overbrace{\hphantom{v_1 \quad v_2 \quad v_3}}^{3} \\ 10\left\{ \begin{pmatrix} v_1 & \vert & v_2 & \vert & v_3 \end{pmatrix} \right. \end{array}$$

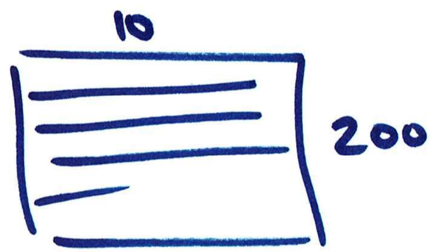
- Scale the data and compute the covariance matrix
- Break the covariance matrix into magnitude and direction. Eigen Vectors and the Eigen Values of the covariance matrix can be thought of as the natural axis/directions and magnitudes along those axis, of the data
    - The eigen values also can be used to calculate the percentage of variation explained by each component
- Sort in the eigen values in desending order and calculate the cumulative percentage of variation explained

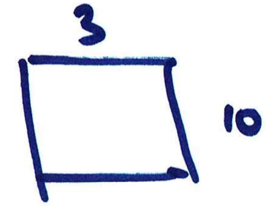
$$\frac{e_1}{\sum e_i}, \frac{e_2}{\sum e_i}, \frac{e_3}{\sum e_i}$$

- Pick the number of principal components you will use
- Transform to new variables

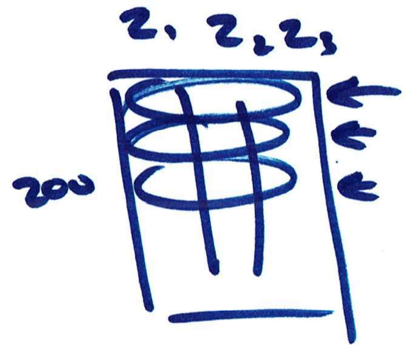




$X =$ 

$W:$ 

$$Z = \underset{200\times 10}{X}\ \underset{10\times 3}{W} \Rightarrow \underset{200\times 3}{Z}$$



$$S = 100 + 20 z_1 + 15 z_2 - 10 z_3$$

$$z_1 : \square x_1 + \square x_2 + \square x_3 + \dots x_{10}$$

$$z_2 : \dots$$

$$z_3 : \dots$$